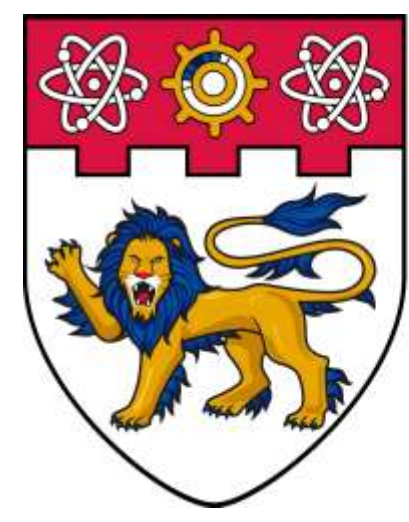




**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Revisiting the Role of Pretrained Weights in Model Merging: On Near-Optimality within the Core Subspace

孙文举 李清勇 李天诚 耿阳李敖 李搏扬



问题设定

模型融合: 将多个微调模型集成为单一模模型, 在不重新训练的情况下, 保持多任务处理能力

任务向量: 考虑一个预训练模型 W_{pre} 以及一系列在下游任务微调的模型 $\{W_i\}_{i=1}^k$ 及下游任务数据集 $\{D_i\}_{i=1}^k$, 各任务向量 $\{\tau_i\}_{i=1}^k$ 定义为 $\tau_i = W_i - W_{pre}$.

Task Arithmetic: 使用缩放系数 λ 将任务向量累加到 W_{pre} , i.e., $W_{mtl} = W_{pre} + \lambda \sum_i \tau_i$, λ 用于控制任务向量累加的强度。

Core Subspace Preservation (CSP)

定义 (核心空间): 对于预训练权重 W_{pre} 其 SVD 分解为 $W_{pre} = U\Sigma V^T$, 其中前 r 奇异向量定义为 U_r 和 V_r . 核心空间定义为

$$\Pi_r(W_{pre}) = \{W | (I - U_r U_r^T)W(I - V_r V_r^T) = 0\}$$

Core Subspace Preservation (CSP):

给定一个 **融合的** 任务向量 τ^* (通过已有方法), CSP 删除了其在核心空间中的部分并融合:

$$\Phi(\tau^*) = (I - U_r U_r^T)\tau^*(I - V_r V_r^T)$$

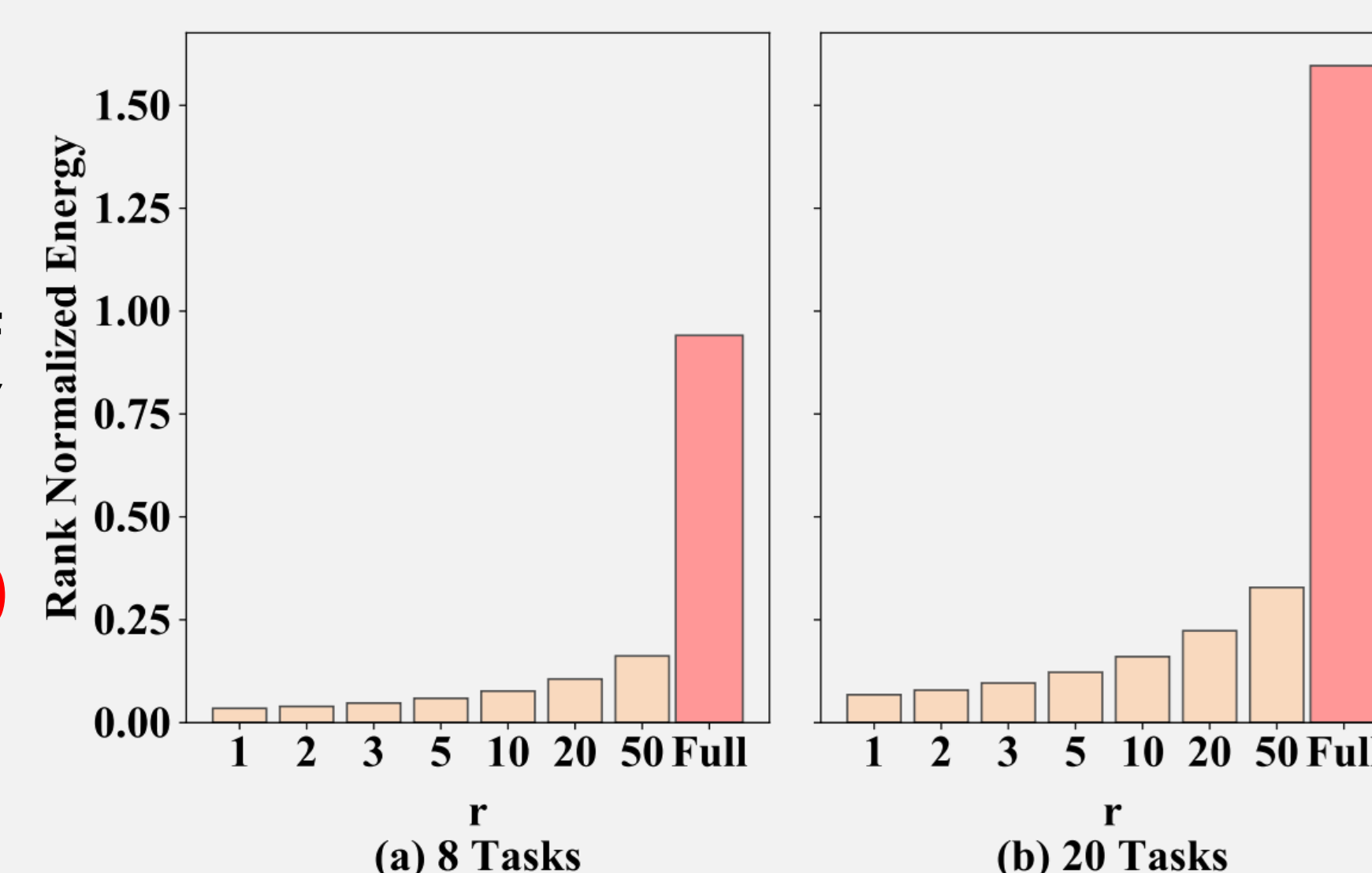
即, **CSP 仅允许 W_{pre} 核心空间外进行模型融合**

W_{pre} 在核心空间接近最优

一阶最优性条件

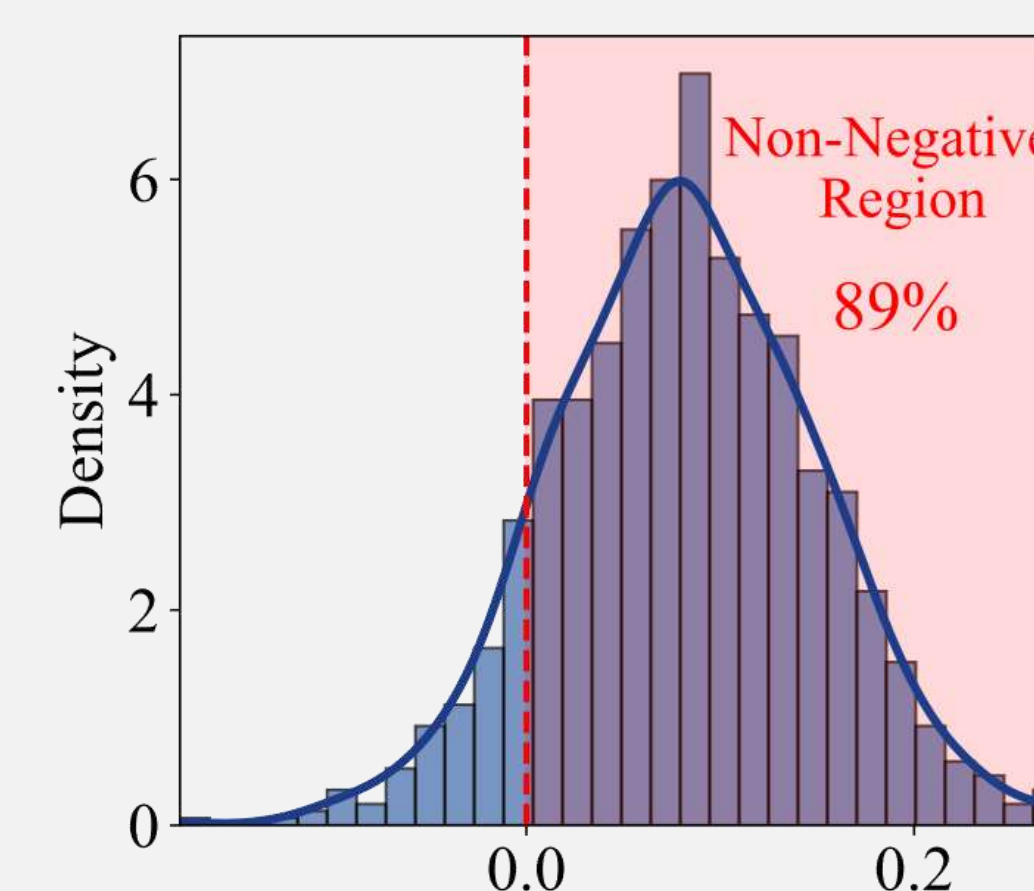
右图表明当 r 比较小时, 梯度趋近于0

W_{pre} 在核心空间处的梯度为0



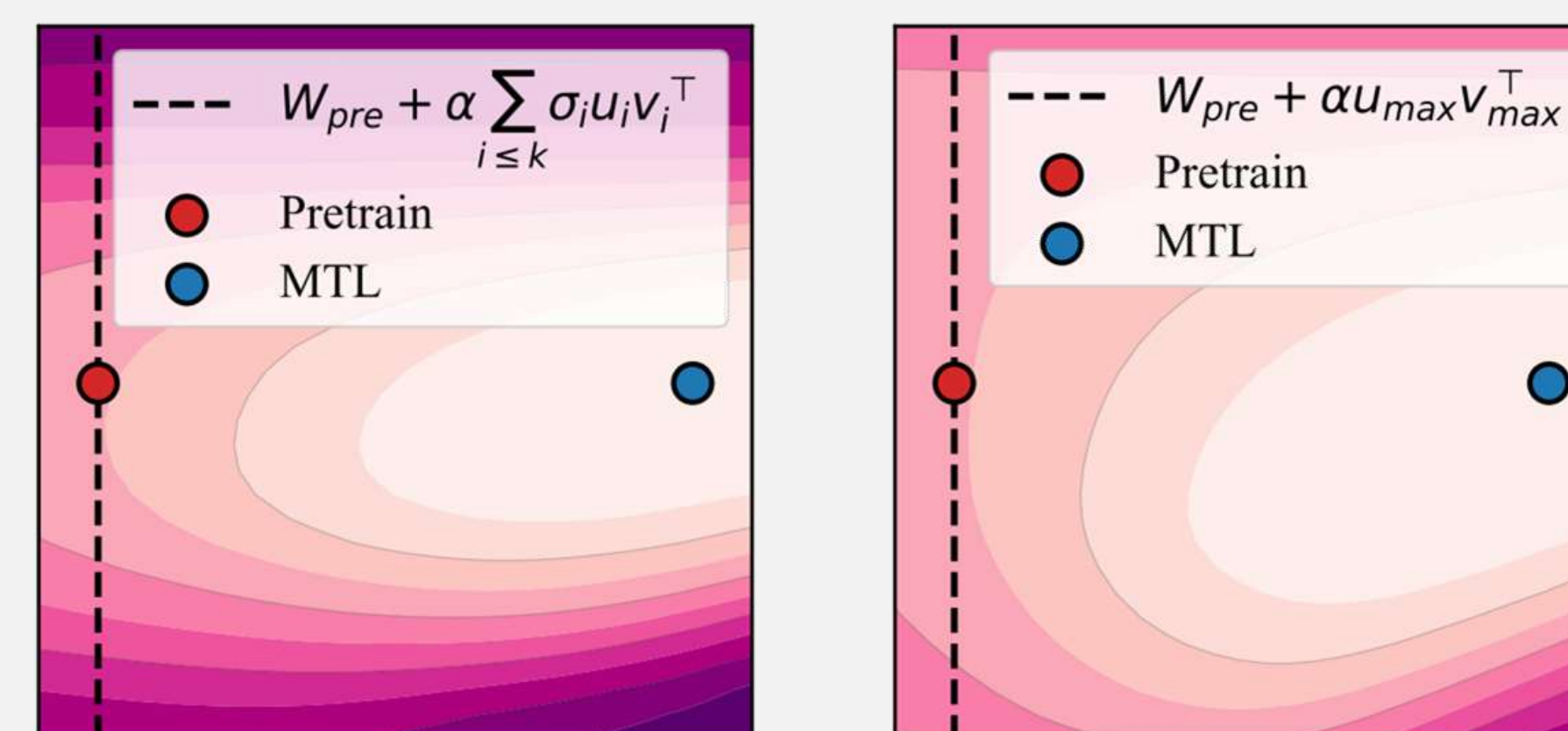
二阶最优性条件

右图表明 89% 随机方向的Hessian非负
核心空间中的曲率保持接近非负



损失景观分析

损失等高线正切于扰动方向(虚线), 表明 W_{pre} 在扰动方向最优



W_{pre} 在核心空间中的信息应当被保护而非融合——CSP

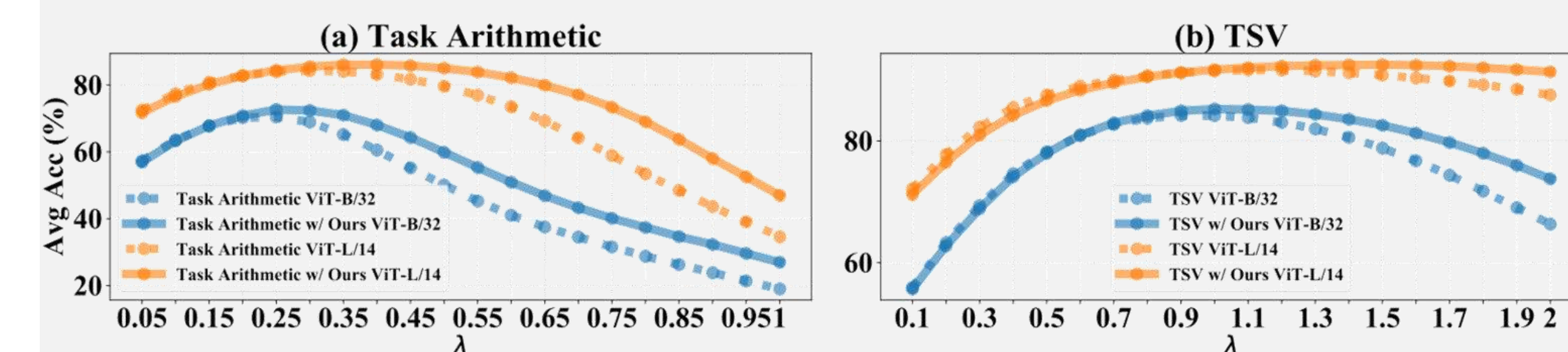
实验结果

融合 8 个 ViT-B/32 模型

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	Avg Acc
pretrained	62.3	59.7	60.7	45.5	31.4	32.6	48.5	43.8	48.0
Individual	75.3	77.7	96.1	99.7	97.5	98.7	99.7	79.4	90.5
Traditional MTL	73.9	74.4	93.9	98.2	95.8	98.9	99.5	77.9	88.9
Task Arithmetic	55.2	54.9	66.7	78.9	80.2	69.7	97.3	50.4	69.1 (75.8)
Task Arithmetic w/ ours	62.9	61.5	71.8	83.4	79.5	70.5	96.8	55.1	72.7 (80.1)
LOT Merging	67.7	67.5	85.7	94.9	93.4	89.8	98.7	63.6	82.7 (90.9)
LOT Merging w/ ours	67.7	66.5	87.0	95.5	94.6	91.6	98.9	67.0	83.6 (91.9)
ISO-C	71.7	70.5	87.0	95.0	90.0	90.8	99.2	68.6	84.1 (92.7)
ISO-C w/ ours	74.2	73.2	87.4	95.4	88.8	91.0	99.1	70.1	84.9 (93.8)
ISO-CTS	74.5	74.1	88.9	93.6	84.7	90.1	98.8	69.8	84.3 (93.2)
ISO-CTS w/ ours	73.7	73.8	88.7	94.7	86.2	90.8	99.0	69.6	84.7 (93.4)
TSV	70.1	72.1	85.9	94.3	90.9	91.2	99.2	68.8	84.1 (92.7)
TSV w/ ours	71.1	73.4	86.3	95.4	92.3	92.6	99.3	70.7	85.1 (93.9)

融合 8 个 ViT-L/14 模型

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	Avg Acc
pretrained	66.8	77.7	71.0	59.9	58.4	50.5	76.3	55.3	64.5
Individual	82.3	92.4	97.4	100.0	98.1	99.2	99.7	84.1	94.2
Traditional MTL	80.8	90.6	96.3	96.3	97.6	99.1	99.6	84.4	93.5
Task Arithmetic	73.9	82.1	86.6	94.1	87.9	86.7	98.9	65.6	84.5 (89.5)
Task Arithmetic w/ ours	75.1	83.6	88.0	95.6	90.2	89.9	99.1	67.8	86.2 (91.3)
LOT Merging	76.7	88.6	91.7	98.7	97.1	95.7	99.5	76.4	90.5 (96.0)
LOT Merging w/ ours	76.3	89.1	93.1	98.9	97.2	96.5	99.6	77.6	91.0 (96.5)
ISO-C	82.9	90.6	95.4	99.0	94.2	97.0	99.4	81.0	92.5 (98.2)
ISO-C w/ ours	83.1	91.4	95.7	99.2	94.0	97.2	99.4	81.5	92.7 (98.5)
ISO-CTS	83.1	91.9	96.0	99.0	93.8	97.9	99.5	82.4	93.0 (98.8)
ISO-CTS w/ ours	83.3	91.8	96.1	99.5	95.1	98.5	99.5	83.0	93.4 (99.2)
TSV	79.2	89.8	93.8	98.9	95.6	96.5	99.5	80.0	91.7 (97.3)
TSV w/ ours	80.5	90.2	94.2	99.4	96.5	97.8	99.7	82.1	92.5 (98.3)



CSP 提升现有方法对 λ 选择的鲁棒性.

联系方式: SunWenJu@bjtu.edu.cn

代码仓库: <https://github.com/SunWenJu123/model-merging>

论文链接: <https://openreview.net/pdf?id=NrGCh79ANM>



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